

Software Requirements Specification (SRS)

Project: Collaborative Study Material & Quiz Platform (CSMQP)

Version: 1.0

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Revision history

Version	Date	Author	Change summary	Approval
1.0	04-09-2026	Bhuvan M S, Bhuvi Dange, Bhuvi Bagga, Bheema Varshini	Initial SRS draft based on project proposal	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose: This document is a Software Requirements Specification (SRS) for the Collaborative Study Material & Quiz Platform (CSMQP). It defines functional and non-functional requirements, interfaces, and verification criteria for the development team, project stakeholders, and assessment evaluators.

1.2 Scope : Covers user registration and authentication, study material sharing, question bank management, quiz creation and attempt, and performance analytics for students and instructors. Excludes integration with external university LMS infrastructure, which is a possible future enhancement.

1.3 Audience: Developers, QA Engineers, Instructors, Students, Administrators, and Assessment Evaluators.

1.4 Definitions: SRS – Software Requirements Specification; RTM – Requirements Traceability Matrix; MERN – MongoDB, Express.js, React.js, Node.js; CRUD – Create, Read, Update, Delete; JWT – JSON Web Token; MCQ – Multiple Choice Question; API – Application Programming Interface; TLS – Transport Layer Security.

2. Overall description

2.1 Product perspective: CSMQP is a new, self-contained web application providing a centralized platform for collaborative academic learning. It operates independently of existing university systems in the initial release.

2.2 Major product functions (detailed)

- User registration, authentication, and role-based access (Student, Instructor, Administrator)
- Study material upload, categorization, search, and download
- Question bank creation and management (MCQ, True/False)
- Quiz creation, publishing, timed attempts, and automatic scoring

- Individual and class-level performance analytics
- Administrative user and content moderation

2.3 User roles and characteristics (expanded)

- Student: basic computer literacy; navigates materials, attempts quizzes, views personal analytics.
- Instructor: basic technical knowledge; uploads materials, manages question bank, creates and publishes quizzes, reviews class performance.
- System Administrator: manages user accounts and moderates platform content.

2.4 Operating environment : Web browser client (desktop, tablet, mobile) over standard internet connectivity; Node.js/Express.js backend; MongoDB database, self-hosted or cloud-hosted.

2.5 Constraints : MERN stack mandatory; JavaScript for both frontend and backend; MongoDB as primary database; uploaded file size limited to 15 MB; must not store financial/payment data.

3. External interface requirements

3.1 User interfaces : Responsive web UI accessible from desktop, tablet, and mobile browsers, with clear navigation between Study Materials, Quizzes, Question Bank, and Analytics sections, and clear success/error feedback.

3.2 Hardware interfaces

No specialized hardware interfaces are required. Standard input/output devices only: keyboard, mouse, touchscreen, display.

3.3 Software interfaces

- Database Interface: backend communicates with MongoDB via Mongoose ODM.
- Authentication Interface: JWT for stateless user authentication.

- API Interface: React frontend communicates with the Node.js/Express.js backend via RESTful APIs.

3.4 Communications : TLS 1.2+ enforced for all client-server communication.

4. System features

Each requirement below includes acceptance criteria and a reference test case. IDs follow CSMQP-F-####.

4.1 User Authentication and Management

Description: Register, authenticate, and manage users with role-based access for Student, Instructor, and Administrator. IDs: CSMQP-F-001..CSMQP-F-003.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQP-F-001	The system shall allow users to register using a valid email address and password.	Functional	High	Student/Instructor	AC-CSMQP-F-001: Registration succeeds only with a valid, unique email and password meeting policy. Test: TC-Auth-01	Password policy TBD
CSMQP-F-002	The system shall authenticate registered users via email and password and issue a JWT session token.	Functional	High	All users	AC-CSMQP-F-002: Valid credentials return a signed JWT; invalid credentials are rejected. Test: TC-Auth-02	Depends on JWT library
CSMQP-F-003	The system shall restrict access to features and API endpoints according to the user's assigned role (Student, Instructor, Administrator).	Functional	High	Security	AC-CSMQP-F-003: Role-mismatched requests return 403. Test: TC-Auth-03	Middleware-enforced

4.2 Study Material Management

Description: Upload, categorize, search, view, and download study materials. IDs: CSMQP-F-010..CSMQP-F-013.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQP-F-010	The system shall allow instructors to upload study materials in PDF, DOCX, PPTX, or JPG formats.	Functional	High	Instructor	AC-CSMQP-F-010: Upload succeeds only for supported formats. Test: TC-Mat-01	File-size limit applies
CSMQP-F-011	The system shall enforce a maximum file size of 15 MB per uploaded material.	Functional	High	Operations	AC-CSMQP-F-011: Files exceeding 15 MB are rejected with an error message. Test: TC-Mat-02	
CSMQP-F-012	The system shall allow study materials to be categorized by subject and topic.	Functional	Medium	Instructor	AC-CSMQP-F-012: Category metadata is stored and displayed with the material. Test: TC-Mat-03	
CSMQP-F-013	The system shall allow students to search and download available study materials by name or keyword.	Functional	High	Student	AC-CSMQP-F-013: Search returns matching materials; download retrieves the correct file. Test: TC-Mat-04	

4.3 Question Bank Management

Description: Create, categorize, edit, and delete questions used in quizzes. IDs: CSMQP-F-020..CSMQP-F-023.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQP-F-020	The system shall allow instructors to add questions to a centralized question bank.	Functional	High	Instructor	AC-CSMQP-F-020: New question is persisted and retrievable. Test: TC-QB-01	
CSMQP-F-021	The system shall support Multiple Choice Questions with question text, four options, and one correct answer, and True/False questions.	Functional	High	Instructor	AC-CSMQP-F-021: MCQ requires exactly one correct option; True/False accepts two options. Test: TC-QB-02	
CSMQP-F-022	The system shall support tagging questions by subject, topic, and difficulty level.	Functional	Medium	Instructor	AC-CSMQP-F-022: Tags are stored and filterable. Test: TC-QB-03	
CSMQP-F-023	The system shall allow instructors to edit or delete existing	Functional	Medium	Instructor	AC-CSMQP-F-023: Edits persist; deleted questions no	Deletion blocked if question is in a published quiz

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	questions.				longer appear in the bank. Test: TC-QB-04	

4.4 Quiz Management

Description: Create, configure, publish, attempt, and score quizzes. IDs: CSMQP-F-030..CSMQP-F-035.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQP-F-030	The system shall allow instructors to create a quiz by selecting questions from the question bank.	Functional	High	Instructor	AC-CSMQP-F-030: Quiz is created referencing selected question IDs. Test: TC-Quiz-01	
CSMQP-F-031	The system shall allow instructors to configure a time limit for each quiz.	Functional	Medium	Instructor	AC-CSMQP-F-031: Configured limit is enforced during attempt. Test: TC-Quiz-02	

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQP-F-032	The system shall allow instructors to publish a quiz, making it visible to eligible students.	Functional	High	Instructor	AC-CSMQP-F-032: Published quiz appears in the student's available list. Test: TC-Quiz-03	
CSMQP-F-033	The system shall allow students to attempt and submit a published quiz within the configured time limit.	Functional	High	Student	AC-CSMQP-F-033: Submission after time limit is rejected or auto-submitted. Test: TC-Quiz-04	
CSMQP-F-034	The system shall automatically calculate the student's score upon submission.	Functional	High	Student	AC-CSMQP-F-034: Score matches the count of correct answers. Test: TC-Quiz-05	
CSMQP-F-035	The system shall display the quiz result, including score and correct/incorrect answers, after submission.	Functional	Medium	Student	AC-CSMQP-F-035: Result view shows per-question correctness and final score. Test: TC-Quiz-06	

4.5 Performance Analytics

Description: Present individual and class-level performance dashboards. IDs: CSMQP-F-040..CSMQP-F-042.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
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Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQP-F-040	The system shall provide students with a dashboard showing average score, recent quiz results, and weak subjects/topics.	Functional	Medium	Student	AC-CSMQP-F-040: Dashboard values match underlying quiz-attempt records. Test: TC-Perf-01	
CSMQP-F-041	The system shall provide instructors with class-level performance statistics, including average scores and question-level performance.	Functional	Medium	Instructor	AC-CSMQP-F-041: Aggregated stats match individual student records. Test: TC-Perf-02	
CSMQP-F-042	The system shall render performance data using charts and graphs.	Functional	Low	Student/Instructor	AC-CSMQP-F-042: Chart renders without error for datasets up to class size. Test: TC-Perf-03	Charting library TBD

4.6 Administration

Description: Manage and moderate user accounts and platform content. IDs: CSMQP-F-050..CSMQP-F-051.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
CSMQP-F-050	The system shall allow administrators to view, block, or delete user accounts.	Functional	High	Administrator	AC-CSMQP-F-050: Blocked/deleted accounts lose login access immediately. Test: TC-Admin-01	
CSMQP-F-051	The system shall allow administrators to moderate and remove reported study materials or questions.	Functional	Medium	Administrator	AC-CSMQP-F-051: Removed content is no longer visible to students/instructors. Test: TC-Admin-02	

5. Non-functional requirements

NFRs below are measurable and tied to test plans. IDs CSMQP-NF-####.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
CSMQP-NF-001	The system shall support up to 500 concurrent users without significant performance degradation.	Performance	High	Load test sustains 500 concurrent sessions. Test: TC-Perf-Load-01
CSMQP-NF-002	Page load times shall not exceed 2 seconds under normal network conditions.	Performance	High	95th percentile page load $\leq$ 2s. Test: TC-Perf-Load-02
CSMQP-NF-003	Quiz submissions shall be processed and graded within 1 second under normal operating conditions.	Performance	Medium	Grading latency $\leq$ 1s in test. Test: TC-Perf-Load-03
CSMQP-NF-004	The system shall maintain 99.5% uptime, excluding scheduled maintenance windows.	Reliability/Availability	High	Monthly uptime reports show $\geq$ 99.5%. Test: Ops reports
CSMQP-NF-005	The responsive interface shall support desktop, tablet, and mobile screen sizes on the latest versions of Chrome, Firefox, Safari, and Edge.	Portability/Usability	Medium	Manual verification across target browsers/devices. Test: TC-UX-01

5.1. Security

5.1.1 Security Objectives

- Protect user credentials and session tokens from disclosure or theft.
- Prevent unauthorized access to, or modification of, another user's data (materials, questions, quizzes, grades).

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
CSMQP-SR-001	User passwords shall be securely hashed (bcrypt or equivalent) before storage; plaintext passwords shall never be persisted.	Security	High	AC: Password field in DB is a salted hash. Test: TC-Sec-01
CSMQP-SR-002	Authentication shall use JWT with a defined expiry; expired or tampered tokens shall be rejected.	Security	High	AC: Requests with expired/invalid JWT return 401. Test: TC-Sec-02
CSMQP-SR-003	Students shall not be permitted to modify quizzes, questions, or grades created by	Security	High	AC: Student-role write attempts on instructor-

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
	instructors.			owned resources return 403. Test: TC-Sec-03
CSMQP-SR-004	The system shall sanitize all user input to prevent Cross-Site Scripting (XSS), Cross-Site Request Forgery (CSRF), and NoSQL injection.	Security	High	AC: Injection payloads in test suite are neutralized. Test: TC-Sec-04
CSMQP-SR-005	All client-server communication shall be encrypted using TLS 1.2 or higher.	Security	High	AC: Server rejects non-TLS connections. Test: TC-Sec-05

6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and RTM shows all test cases passed.
- Acceptance test suites: Authentication, Study Materials, Question Bank, Quiz, Performance Analytics, and Security tests.

7. System models and diagrams

7.1 UML Use-Case diagram

Use-case diagrams for (a) Student interactions (view/search/download materials, attempt quiz, view analytics) and (b) Instructor interactions (upload materials, manage question bank, create/publish quiz, view class analytics) to be added.

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
CSMQP-F-001	User registration	4.1 / Auth Module	AuthModule	TC-Auth-01	N	
CSMQP-F-010	Upload study material	4.2 / Material Module	MaterialModule	TC-Mat-01	N	
CSMQP-F-030	Create quiz	4.4 / Quiz Module	QuizModule	TC-Quiz-01	N	
CSMQP-F-033	Attempt quiz	4.4 / Quiz Module	QuizModule	TC-Quiz-04	N	
CSMQP-F-040	Student performance dashboard	4.5 / Analytics Module	AnalyticsModule	TC-Perf-01	N	
CSMQP-	Concurrent user	5 / Infra	WebUI / API	TC-Perf-	N	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
NF-001	support			Load-01		
CSMQP-SR-001	Password hashing	5.1.2 / Auth Module	AuthModule	TC-Sec-01	N	